Glossary of Key Terms

Structuring Machine Learning Projects

AI Foundations Series — Book 3

This glossary collects the main terms and concepts from Book 3 — the strategy of real machine-learning work — arranged alphabetically for quick reference. Each definition reflects how the term is used in the book itself: plain language first, with just enough precision to be useful when reviewing a chapter or returning to one of the four companion programs. Beginners can read it straight through; reviewers can jump to any entry.

A

Accuracy

The fraction of predictions a model gets right. Simple and intuitive, but it can mislead on imbalanced problems (a 99%-cats dataset rewards a model that always guesses “cat”), which is why precision, recall, and F1 exist.

Avoidable Bias

The gap between human-level (or Bayes) error and the model's training error. It is the part of the training error you could realistically hope to remove. A large avoidable bias points to an underfitting problem - train longer, use a bigger model, or a better architecture.

B

Bayes Error

The lowest error rate any classifier could possibly achieve on a task - the irreducible noise in the problem itself. It can never be beaten, only approached, and it serves as the ultimate yardstick for how much room a model has left to improve.

D

Data Augmentation

Expanding a dataset by adding modified copies of existing examples (for instance, adding background noise to audio). Useful for closing a data-mismatch gap, but risky if synthesized from too small a pool, since the model may overfit the narrow set of artificial variations.

Data Mismatch

When the training data is drawn from a different distribution than the dev and test sets, so the model does well on what it trained on but poorly on what it is actually graded on. It looks like high variance but is a distribution problem, and the training-dev set is what tells them apart.

Dev Set (Development Set)

Held-out data used to tune choices and measure progress while building the model. The one firm rule: the dev and test sets must come from the same distribution and reflect the data you truly care about - they set the target everyone aims at.

Distribution

The underlying source and character of a dataset - where the examples come from and what they look like. Many of this book's hardest problems come down to training and testing data being drawn from different distributions.

E

End-to-End Deep Learning

Replacing a hand-built pipeline of separate stages with a single network that maps raw input directly to the final output. It can outperform a pipeline by letting the data speak, but it is data-hungry, so a staged pipeline still wins when labeled data is scarce.

Error Analysis

Manually examining a sample of a model's mistakes - often a tally of about a hundred mis-classified dev examples - to decide where fixing effort will pay off most. It turns “the model is wrong sometimes” into a ranked list of concrete problems to attack.

F

F1 Score

A single number that folds precision and recall together (their harmonic mean), so two competing metrics become one you can optimize. Useful because a model usually cannot maximize precision and recall at the same time.

H

Human-Level Performance

How well people do on a task, used as a practical stand-in for Bayes error. Comparing a model against it reveals how much avoidable bias remains - and once a model surpasses human level, that yardstick stops being reliable.

M

Multi-Task Learning

Training one network to do several related tasks at once (for example, a self-driving car detecting cars, pedestrians, and signs from the same image), so shared features help every task. It also lets partially labeled examples still contribute.

O

Optimizing Metric

The single metric you try to make as good as possible - the number you are genuinely maximizing or minimizing, such as accuracy. Paired with one or more satisficing metrics when a problem has more than one thing you care about.

Orthogonalization

The organizing principle of the book: arrange your work so each “knob” adjusts one thing without disturbing the others, the way orthogonal (right-angle) directions are independent. Independent knobs make it possible to diagnose and fix problems one at a time.

P

Precision

Of all the examples the model labeled positive, the fraction that truly were positive. It answers “when the model says yes, how often is it right?” and trades off against recall.

R

Recall

Of all the examples that were truly positive, the fraction the model caught. It answers “of everything it should have found, how much did it find?” and trades off against precision.

S

Satisficing Metric

A metric that only has to clear a bar rather than be maximized - below the bar it is a deal-breaker, above the bar you stop caring. Combined with an optimizing metric, it resolves trade-off arguments (e.g. “maximize accuracy, as long as it runs under 100 ms”).

Single-Number Evaluation Metric

One metric that summarizes performance, so two models can be compared at a glance and progress is unambiguous. Having a single number to chase dramatically speeds up the build-measure-improve loop.

T

Test Set

Data held back and used only at the very end to estimate real-world performance honestly. It must share a distribution with the dev set and is never tuned against.

Training Set

The data the model actually learns from by adjusting its weights. In this book it may deliberately be drawn from a larger or different distribution than the dev and test sets.

Training-Dev Set

A slice held out from the training data but drawn from the same distribution as the training set. Comparing errors across train, training-dev, dev, and test pinpoints whether a problem is avoidable bias, variance, or data mismatch.

Transfer Learning

Reusing what a network already learned on one task to give it a head start on another - for example, taking a model trained on everyday photos and adapting it to diagnose plant disease. Most valuable when the new task has little labeled data.

V

Variance

The gap between a model's training error and its dev error. A large variance means the model fits its training data far better than new data - it is overfitting. The fix is more data, regularization, or a simpler model.

W

Workflow (ML Strategy Loop)

The disciplined cycle this book teaches: aim at a measurable target, diagnose before fixing, find the biggest wins with error analysis, tell variance from data mismatch, and choose a learning approach to match your data - then iterate.